User Guidelines for the Map of EU Political Information

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Based on DELIVERABLES 1.4 and 1.6

MeDeMAP – Mapping Media for Future Democracies

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1. Introduction

MeDeMAP endeavors to communicate research results to all those interested in the relationship between media and democracy, be they politicians, representatives of media authorities and self-regulation bodies, media owners, journalists, NGOs, or citizens, who are most affected by how this relationship is shaped. Therefore, an interactive country mapping tool of EU political information environments has been developed, which is openly accessible to everyone.

The idea behind this tool is to help analyse to what extent and in what ways legal and political framework conditions, media supply, and media use are interconnected in EU Member States, and what this means for the state of democracy and political participation in these countries. This tool enables critical comparison of the characteristics of several indicators across all EU member states, thereby recognising possible relationships among the media law and regulatory framework conditions, the composition and quality of media offerings, the use of and trust in different media, and different notions of democracy and participation.

In pursuing this objective, **'Map of EU Political Information Environments'** does not simply provide some figures; it

- includes metadata such as the original definition, collection method, and calculation mode;
- provides data for similarly defined (but perhaps differently labeled) indicators from different sources and for similarly labeled indicators following different approaches, because a more reliable judgement on the plausibility of data can often only be made by comparison;
- presents re-coded data to make them comparable and calculated statistical measures that describe the structure of data and allow for defining threshold values that give an initial orientation of how a country behaves in comparison to other countries;
- includes data for all indicators covering the entire project period (2022 – 2024), which allows comparisons to be made, at least for a short but politically turbulent time period.

To support critical data analysis, "mapping" is understood as a process of creating a structured visual or digital representation of data to improve understanding and decision-making. This means that, in addition to the geographical representation of data, users of the "Map of EU Political Information Environments" are offered a range of different formats that encourage creative work with the data.

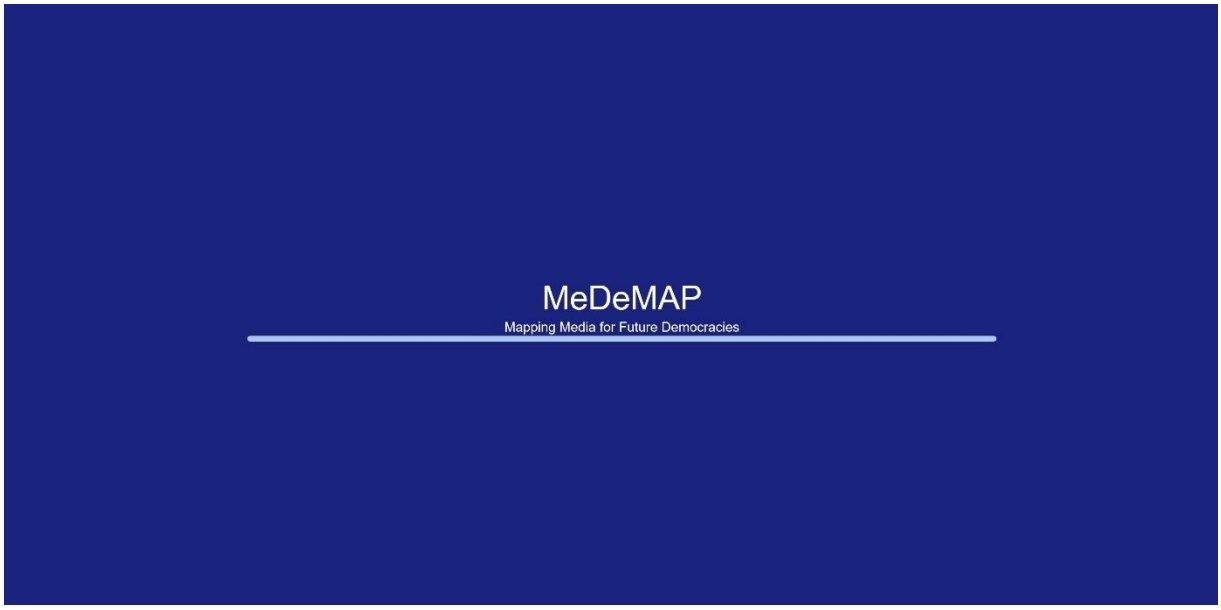
The “Map of EU Political Information Environments” is subject to the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0), which allows users to share and adapt the content for non-commercial purposes, provided proper attribution is given. This license ensures that users can freely use the data while respecting the terms outlined by the license. The “Map of EU Political Information Environments” re-uses data from various data providers. If this data has not been published under a Creative Commons license, written permission from the data provider to re-use the data has been obtained. Users are encouraged to refer to the Data Annotations for detailed information on acknowledgments and specific terms of use related to external data sources integrated into this site.

For proper use of the “Map”, users are required to cite the following paper in any publications or works that use the data obtained from this site:

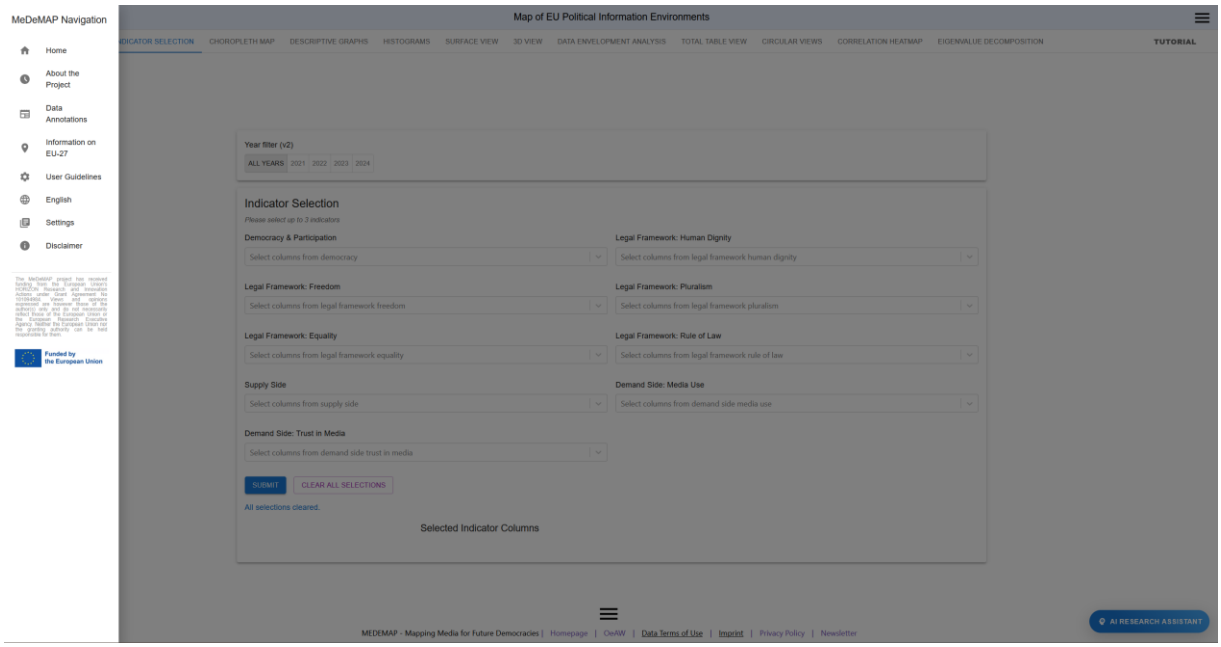
Seethaler, J., Beaufort, M., & Lentz, T., with contributions by Beata Klimkiewicz and Panos Kompatsiaris (2026). Map of EU Political Information Environments (MeDeMAP – Grant Agreement No 101094984). Vienna: OEAW. https://medemap.cmclab.science/

2. Structure of the Online Tool

After accessing the website via https://medemap.cmclab.science/, this **loading screen** appears during the very short loading time of the tool:



The tool then starts with the **menu bar** on the left-hand side, which can be hidden and shown again using the menu icon ☰ at the bottom or in the right upper corner of the website. The Escape key also makes the menu disappear.



The menu allows for accessing some essential information **About the Project**, extensive **Data Annotations** and **Information on the EU-27 Member States**. Some settings can also be changed here (such as the 'Indicator Label Format'), but this requires a more experienced user.

Since "Mapping" is understood as a process of creating a structured visual or digital representation of data to improve understanding and decision-making, the users of the "Map of EU Political Information Environments" are offered a range of different formats that encourage creative work with the data. These formats are:

Starting point: Indicator selection (all EU-27)	Going deeper: Alternative selections of indicators and countries	All data
• Choropleth map • Descriptive graphs • Histograms • Surface view • 3D view	• 3D view • Data Envelopment Analysis (DEA) • Total table view • Circular views	• Total table view • Circular views • Eigenvalue decomposition • Correlation heatmap

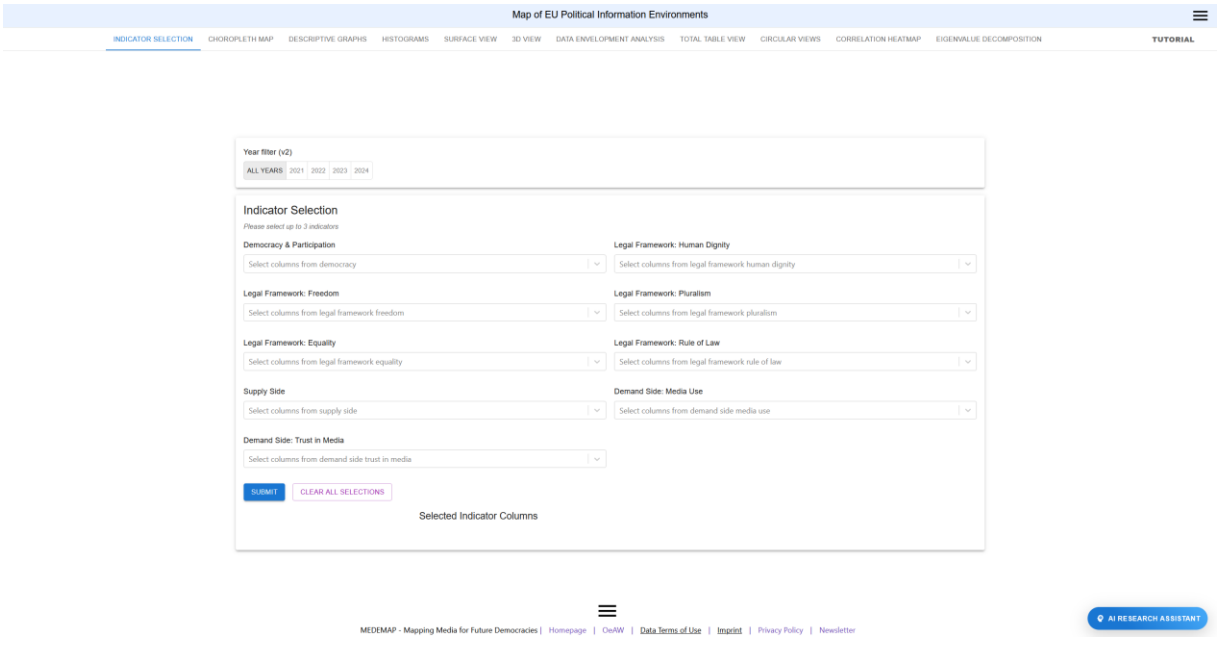
The different formats not only provide different representations of the data but also enable different approaches to working with it:

- The starting point—recommended for getting started with the map—begins on the "Indicator Selection" subpage: the data for the indicators selected there are visualized in various formats.
- Other formats offer analysis options that can be used independently of these selected indicators and are more flexible.
- Some other formats are based on all data; here, the "Total Table View" in particular offers a complete overview of the database.

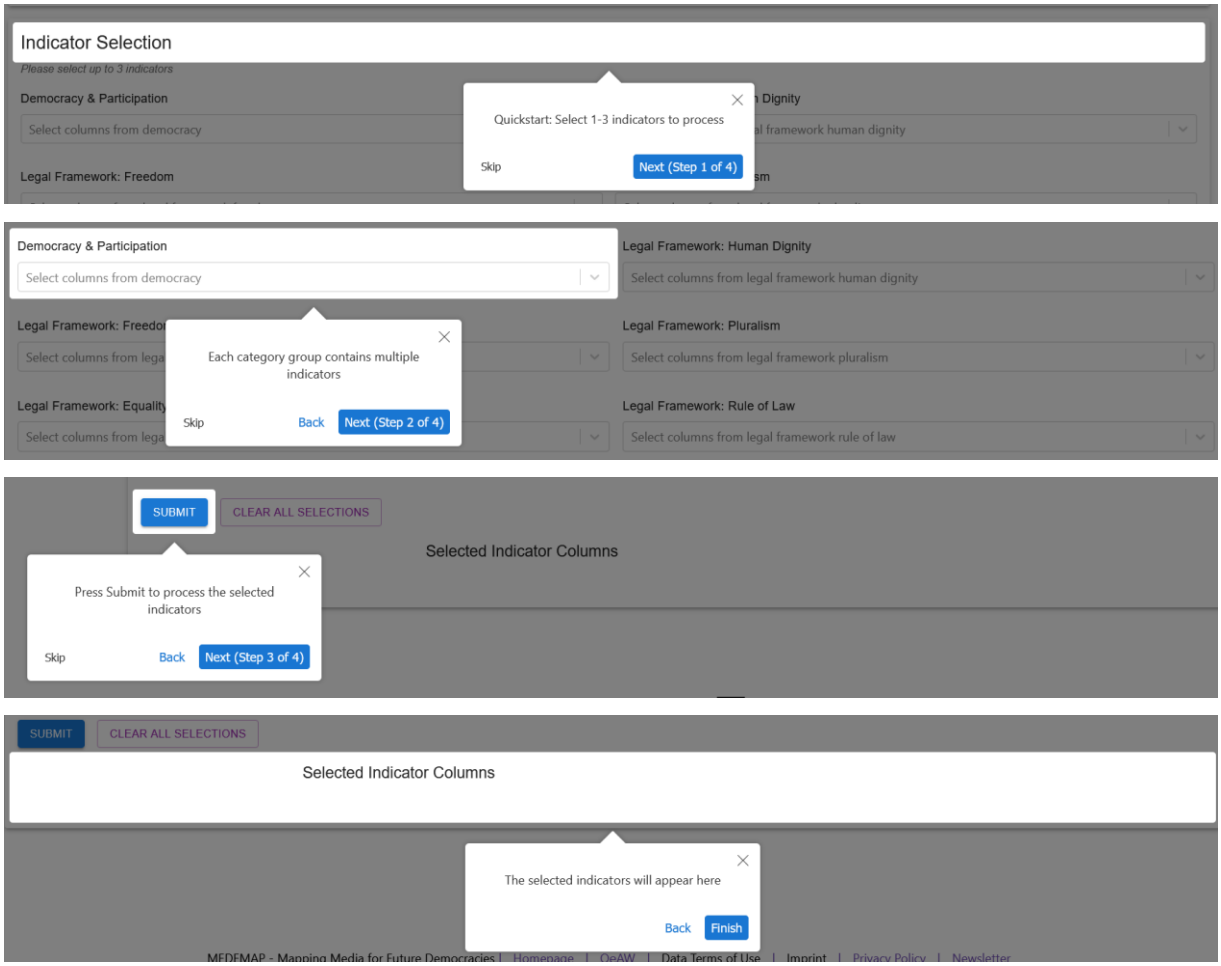
2.1. Starting Point: Indicator Selection (all EU-27)

2.1.1. Indicator Selection

Having clicked away the menu bar, the **Indicator Selection** page appears.

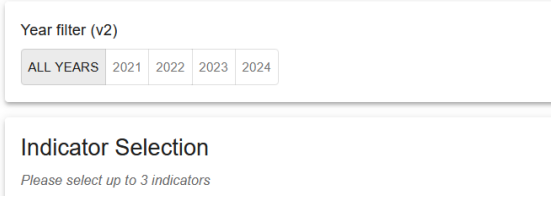


On the right side, at the top of the page, a short but useful **Tutorial** can be started. It guides the user through the selection process:



The user may select **up to three indicators** from any indicator sector for visualisation and can clear the selection at any time (using the 'CLEAR ALL SELECTIONS' button) to create a new one. After making a selection, the **selected indicators must be submitted** ('SUBMIT' button).

It is possible to select three indicators from one section, or one indicator from three sections, or one indicator from one and two indicators from another section. Selecting three indicators from a single section is particularly appropriate when changes over time are to be examined. If you want to compare data from several indicators for a specific year, it is advisable to select the year of interest in the year menu before selecting the indicators.



The year refers to the data's validity year. Please note that data for a specific year are usually published in the following year. (That means that the data contained in the 2023 to 2025 editions

of a certain data source usually applies to the years 2022 to 2024. However, in most cases, the title of the relevant publications is misleading, as it mentions the year of publication while the data refers to the previous year. For more information see the "Data Annotations".)

Only in two cases, due to the available data, 2021 (instead of 2022) was chosen as the starting year of the map's data collection.

Based on MeDeMAP research, the database currently encompasses the following sections:

- Democracy & Participation
- Legal Framework: Human Dignity
- Legal Framework: Freedom
- Legal Framework: Pluralism
- Legal Framework: Equality
- Legal Framework: Rule of Law
- Supply Side
- Demand Side: Media Use
- Demand Side. Trust in Media

The available formats of visualisation can be selected in the **header**:



2.1.2. Choropleth Maps

The **Choropleth Maps** show how a quantifiable variable varies across geographic areas, using different colours to represent data values across the 27 EU Member States. They use a traffic-light colour scheme, with green for high values, orange for medium values, and red for low values, making it easy to visualize spatial patterns and variations at a glance. Here's the key to explaining the color-data relationship:

Data level	Threshold value	Color
High	*Above* mean *plus* half a standard deviation ($X > \mu + \sigma / 2$) If source = Economist Intelligence Unit – Democracy Index: data ≥ 80.00 If source = Media Pluralism Monitor: data ≥ 66.67	○ **green (comparatively 'best' or 'most')**
Medium	*Between* mean *plus* half a standard deviation *and* mean *minus* half a standard deviation ($\mu + \sigma/2 > X > \mu - \sigma/2$) If source = Economist Intelligence Unit – Democracy Index: data < 80.00 and ≥ 60.00 If source = Media Pluralism Monitor: data < 66.67 and > 33.33	○ **orange ('somewhere in the middle')**
Low	*Below* mean *minus* half a standard deviation ($X < \mu - \sigma / 2$) If source = Economist Intelligence Unit – Democracy Index: data < 60.00 and ≥ 40.00 If source = Media Pluralism Monitor: data ≤ 33.33	○ **red (comparatively 'worst' or 'least')**

Where possible, the threshold values are determined based on theoretically substantiated information from the sources. If this information is not available (as is usually the case), a widely used measure in statistics for describing data series is used, which adds half a standard deviation to the mean value for the threshold value between the upper and middle data range and subtracts half a standard deviation from the mean value for the threshold value between the middle and lower data range. This procedure is not comparable to a theoretically sound definition of thresholds, but it does provide an initial orientation to how a country behaves relative to other countries.

2.1.3. Descriptive Graphs

The **Descriptive Graph View** allows for an easy comparison of the data of the selected indicators by displaying three bar charts. While again using a traffic-light colour scheme, with green for high values, orange for medium values, and red for low values, the graphs highlight differences across countries, making it easier to see patterns and trends.

2.1.4. Histograms

The **Histograms** show the distribution of the data according to consecutive, non-overlapping intervals ('bins'), thus, for example, allowing 'outliers' and data value clusters to be recognised at a glance. Histograms reveal patterns hidden within unordered data and exhibit an organized picture that makes interpretation and analysis much more manageable and effective. The number of bins, the colour mode, and the colour range can be changed. The default setting is dark blue for high values, mid-blue for medium values, and light blue for low values. However, for mathematical reasons, the boundaries between the value ranges cannot be drawn as strictly as in choropleth maps and descriptive graphs.

2.1.5. Surface View

The **Surface View** displays 3D surface plots, allowing users to rotate, zoom, and translate the view for detailed analysis. The plot can be downloaded as a png file.

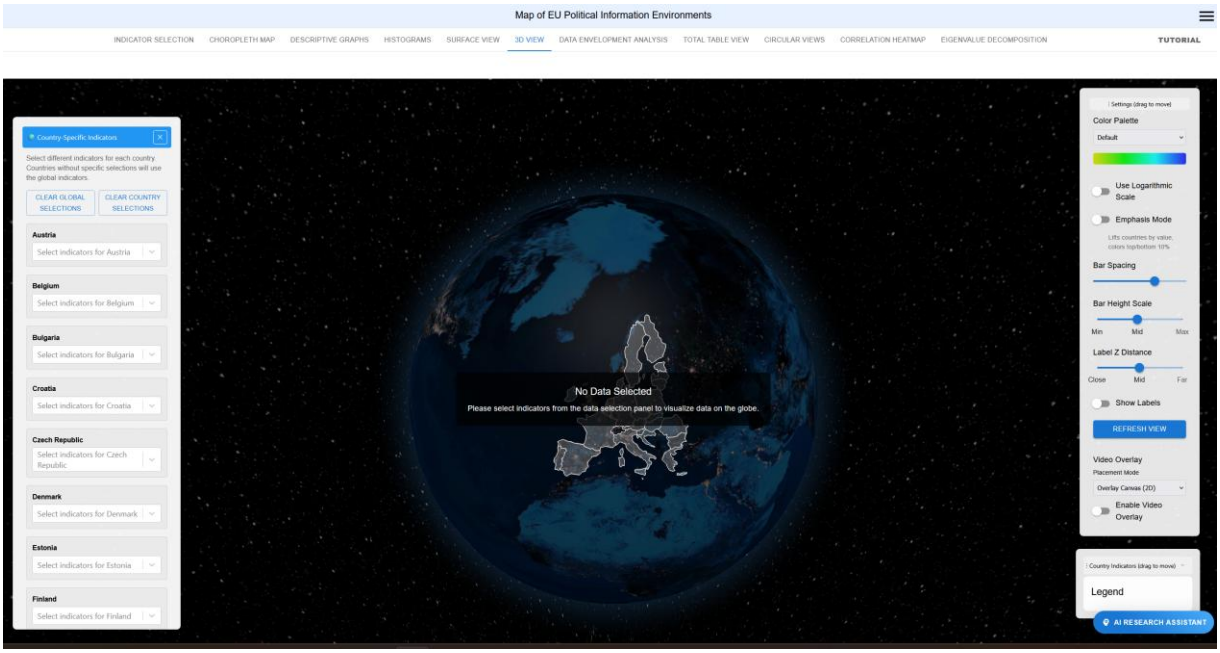
2.1.6. 3D View (1)

The **3D View** visualises data values on the surface of a globe, with the Member States of the European Union highlighted. A number of features can be tweaked: the colour palette, bar spacing, bar height, showing of (text and numeric) labels, label Z distance (= distance of label in

relation to the bar), label size. If the default settings are changed, refreshing the view is recommended (button 'REFRESH VIEW'). The menu with these settings is located on the right side of the screen and can be moved by dragging it.

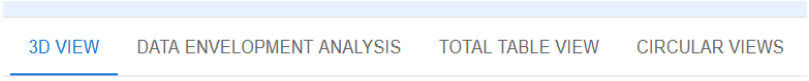
In case only one indicator has been selected, The 'EMPHASIS MODE' highlights the 10 percent of countries with the highest (green) and lowest values (red).

A special feature is the option to embed any video.



2.2. Going deeper: Alternative selections of indicators and countries

Again, the available formats can be selected in the **header**:



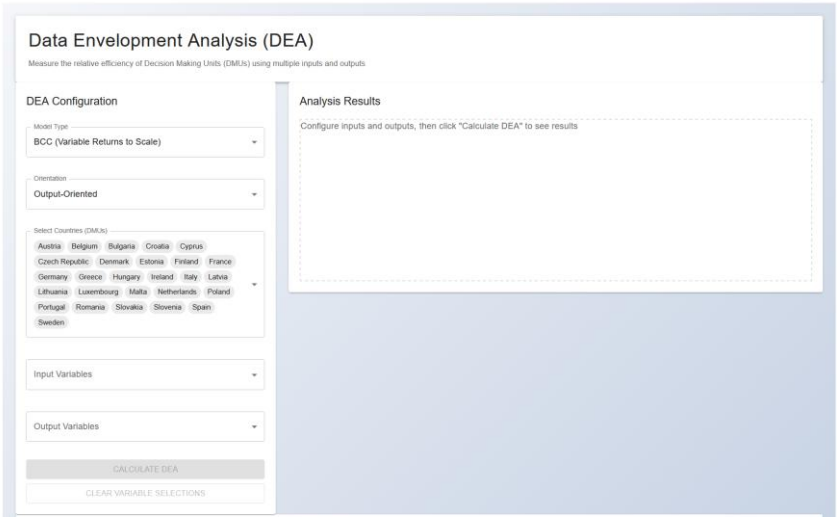
2.2.1. 3D View (2)

The **3D View** offers an additional selection mode ('COUNTRY-SPECIFIC INDICATORS'), which allows any number of specific indicators to be selected for each country and then compared with the same or other indicators across any number of other countries. The menu with these selection mode is located on the left side of the screen and can be clicked away.

2.2.2. Data Envelopment Analysis

Data Envelopment Analysis (DEA) is a nonparametric technique using linear programming to measure the *relative* efficiency of similar Decision-Making Units (DMUs) with multiple resources (inputs) and results (outputs). The efficiency score ranges from 0% to 100%, with 100% meaning that the DMUs are at the efficiency frontier. This implementation uses the output-oriented BCC model to assess the relative efficiency of European countries with regard to selected media and democracy indicators. It handles multiple inputs and outputs simultaneously, and identifies specific areas for improvement. It is a method that sharpens relational thinking and assessment, but does not specify any content-related objectives – the analyst must keep these goals in mind when analyzing units that tend to be inefficient in order to determine where improvements are needed.

The implemented DEA tool allows selection of countries (the default setting is EU-27), input variables, and output variables. The analysis results can be exported as csv file.



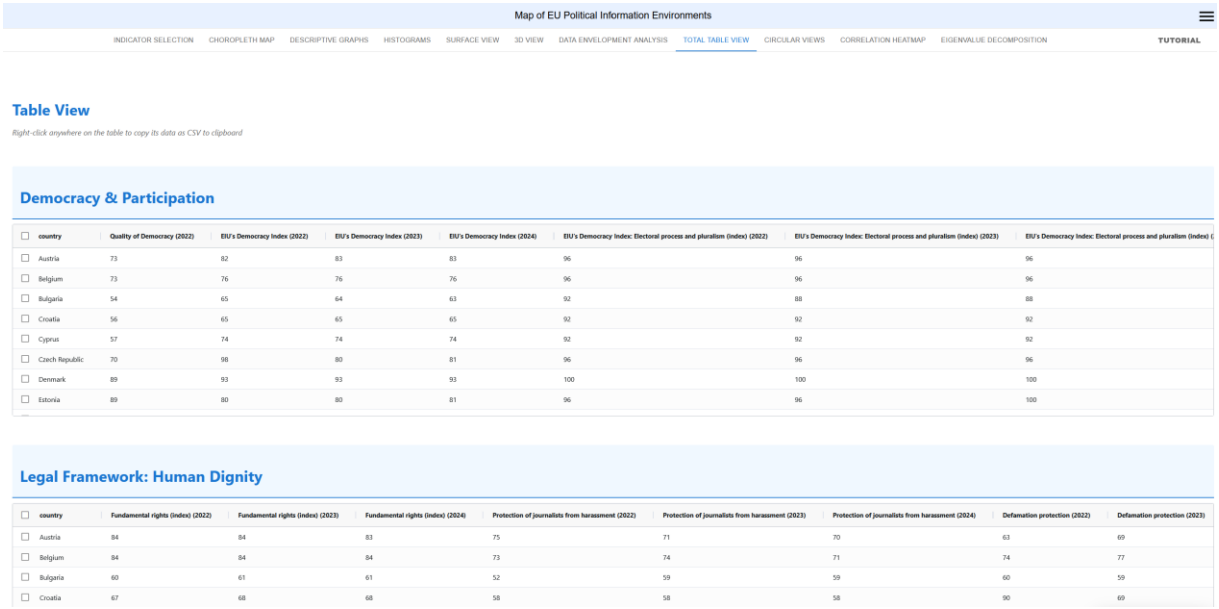
2.2.3. Total Table View

Another special feature of the 'Map of EU Political Information Environments' is the **Total Table View**. It presents all the data fed into the database. The tool thus fully complies with Horizon Europe's Open Access policy.

The data is arranged along the nine sections listed in Chapter 2.1.1. The data in each section can be sorted in ascending or descending order by country or by the values of each indicator, enabling quick and direct comparison of all data in a section. Any selection of countries can be made.

A right-click anywhere on a table containing data for a section allows copying its data as a CSV to the clipboard or exporting it as a CSV file. Here's how it works:

- Copy CSV Table to the Clipboard → Select rows and choose 'Copy CSV Table to the Clipboard' → Paste data into an Excel Sheet → Click on Paste Options: Text Import Wizard → Choose "Delimited" and "My data has headers" → NEXT: Delimiters: Choose ONLY 'Comma' and Text Qualifier: 'none' → NEXT: Column data format: General; Click on 'Advance' and choose your favourite Decimal separator → FINISH
- Export Table as CSV file: save csv file → open with Excel: Data → Get data → From File → From Text/csv → Choose the downloaded csv file → File origin: Western European OR Unicode (UTF-8); Delimiter: Comma; Data Type Detection: 'do not detect data type' → Load



2.2.4. Circular View

While the 3D View and the Total Table View allow for a quick comparison of the values of different or all indicators in a section for a selected country or any selection of countries, the **Circular View** provides – after an overview based on all indicators – an impressive visualisation of the data for any selected indicator. All it takes is to click on the desired indicator from a list of all indicators. The plot can be downloaded as a png file.

2.3. All data

Again, the available formats can be selected in the **header**:

Besides the Total Table View (see Chapter 2.2.3) and the Circular View (see Chapter 2.2.4), the **Correlation Heatmap** and the **Eigenvalue Decomposition** are visual representations of the entire database.